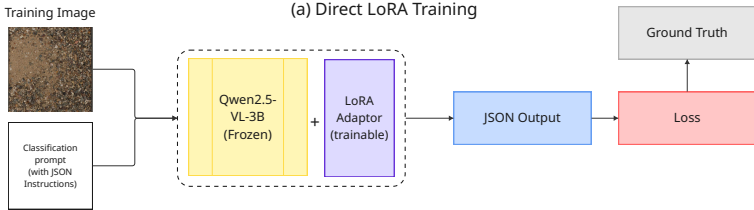


(a) Direct LoRA Training



N training pairs (1 per image, 18-30 total)

Classification Prompt with JSON Instruction at the End

Classify this concrete aggregate photograph.
Ground sampling distance (GSD) = 2.1 px/mm.
At this GSD: 8mm $\approx$ 17px, 16mm $\approx$ 34px, 32mm $\approx$ 68px.

Classification axes:

1. MAX PARTICLE SIZE: estimate the largest stone's width in pixels, divide by GSD, round to 8, 16, or 32 mm.
2. GRADING (DIN 1045 standard):
 - COARSE (A): gaps between stones EMPTY
 - MEDIUM (B): gaps PARTIALLY filled
 - FINE (C): gaps COMPLETELY filled

Respond with JSON: {"max_particle_size_mm": <8|16|32>, "grading": ""}

CoT Descriptions

The largest stones in this image appear to be around 16mm, fitting the max_particle_size_mm: 16 classification. The gaps between the largest stones are mostly empty, with very few smaller particles filling the spaces, which is characteristic of a coarse grading. The overall texture looks uniform and not densely packed with mixed sizes, supporting the classification of coarse aggregate according to DIN 1045.

CoT Description Generation Prompt

Look at this concrete aggregate photograph. The correct classification is:
max_particle_size_mm: 16, grading: coarse

Ground sampling distance (GSD) = 2.1 px/mm.
At this GSD: 8mm stone $\approx$ 17px, 16mm $\approx$ 34px, 32mm $\approx$ 68px.

Grading definitions (DIN 1045 standard — describes size DISTRIBUTION):

- COARSE (A): particles concentrated near max size. Gaps between stones are EMPTY. Uniform, single-layer texture.
- MEDIUM (B): balanced mix. Gaps PARTIALLY filled by smaller particles.
- FINE (C): wide size range. Gaps COMPLETELY filled with small particles. Dense, packed texture.

Key distinguishing features:

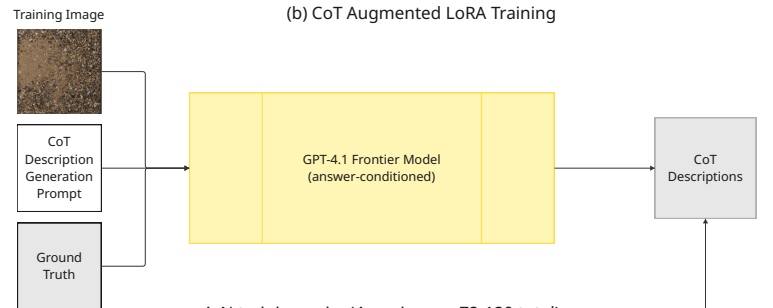
- coarse vs medium: coarse has EMPTY gaps (you can see the background), medium has gaps PARTIALLY filled with smaller stones
- medium vs fine: medium still has some visible gaps, fine has NO visible gaps (everything packed tight)
- 8mm vs 16mm vs 32mm: estimate the largest stone's width in pixels,

JSON Output

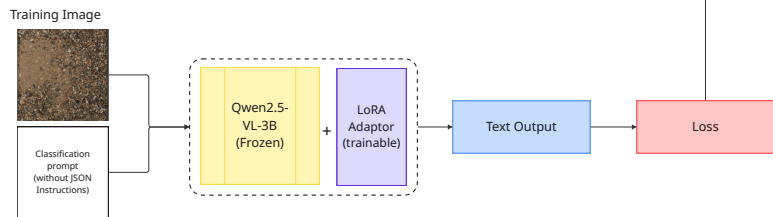
{"max_particle_size_mm": 16, "grading": "coarse"}

Result: Base 12.0% $\rightarrow$ Direct LoRA 71.3% $\rightarrow$ CoT-Augmented 79.6% (+8.3pp)

(b) CoT Augmented LoRA Training



4xN training pairs (4 per image, 72-120 total)



Training Image

